Basic Packet Capture

tcpdump

tcpdump is a command-line packet analyzer used to capture and analyze network traffic.

Common uses:

- Network troubleshooting
- Security analysis
- Packet inspection
- Capturing traffic for Wireshark

Basic syntax:

```
tcpdump [options]
```

Specify Network Interface (-i)

You must tell tcpdump which network interface to capture packets from.

Command:

```
tcpdump -i <interface>
```

Example:

```
tcpdump -i eth0
```

Capture packets from eth0 interface.

Capture from all interfaces:

```
tcpdump -i any
```

Save Packets to File (-w)

To store captured traffic for later analysis.

Command:

```
tcpdump -w file.pcap
```

Example:

```
tcpdump -i eth0 -w capture.pcap
```

Read Packets from File (-r)

Read a previously captured file.

Command:

```
tcpdump -r file.pcap
```

Example:

```
tcpdump -r capture.pcap
```

Limit Number of Packets (-c)

By default, tcpdump runs until interrupted (CTRL + C).

To limit capture:

```
tcpdump -c <count>
```

Example:

```
tcpdump -i eth0 -c 10
```

Capture only 10 packets.

Disable DNS Resolution (-n)

Normally tcpdump converts:

```
IP → domain name
```

Example:

```
172.217.1.1 → google.com
```

To disable this:

```
tcpdump -n
```

Example:

```
tcpdump -i ens5 -n
```

Disable Port Resolution (-nn)

Tcpdump also converts ports:
80 → http
22 → ssh

Disable both DNS and port resolution:

```
tcpdump -nn
```

Example:

```
tcpdump -i ens5 -nn
```

Output example:

```
10.10.117.2.22 > 10.11.81.126.53378
```

Verbose Output (-v)

Shows more packet details.

Levels:

-v	Verbose
-vv	More verbose
-vvv	Maximum verbosity

Example:

```
tcpdump -i eth0 -v
```

Shows additional details like:

- TTL
- Packet length
- Protocol information

Common Practical Examples

1. Capture 50 packets with details

```
tcpdump -i eth0 -c 50 -v
```

Meaning:
- Interface: eth0
- Packets: 50
- Verbose output

2. Capture packets and save them

```
tcpdump -i wlo1 -w data.pcap
```

Meaning:
- Interface: WiFi
- Save packets to data.pcap
- Stop using CTRL + C

tcpdump Filtering

Capturing all packets is not useful because networks generate thousands of packets per second.

Filtering helps you:
- Focus on specific traffic
- Reduce noise
- Analyze attacks faster

Filtering by Host

You can filter packets related to a specific IP address or hostname.

Basic Syntax

```
tcpdump host <IP or HOSTNAME>
```

Example:

```
sudo tcpdump host example.com
```

Source Host Filter

Capture packets coming from a specific host.

```
tcpdump src host <IP>
```

Example:

```
tcpdump src host 192.168.1.5
```

Captures packets sent from 192.168.1.5.

Destination Host Filter

Capture packets going to a specific host.

```
tcpdump dst host <IP>
```

Example:

```
tcpdump dst host 192.168.1.5
```

Captures packets sent to 192.168.1.5.

Filtering by Port

Many protocols use specific port numbers.

Protocol	Port
HTTP	80
HTTPS	443
DNS	53
SSH	22
NTP	123

Basic Port Filter

```
tcpdump port <PORT_NUMBER>
```

Example:

```
tcpdump port 53
```

Captures DNS traffic.

Source Port Filter

```
tcpdump src port <PORT>
```

Example:

```
tcpdump src port 53
```

Captures packets originating from port 53.

Destination Port Filter

```
tcpdump dst port <PORT>
```

Example:

```
tcpdump dst port 80
```

Captures packets going to HTTP servers.

Filtering by Protocol

You can filter traffic by network protocol.

ip	IPv4 packets
ip6	IPv6 packets
tcp	TCP traffic
udp	UDP traffic
icmp	Ping traffic

Example ICMP Filter

```
tcpdump icmp
```

Logical Operators

AND Operator

Captures packets where both conditions are true.

Syntax:

```
tcpdump condition1 and condition2
```

Example:

```
tcpdump host 1.1.1.1 and tcp
```

Captures:

- TCP packets
- involving host 1.1.1.1

OR Operator

Captures packets where either condition is true.

Syntax:

```
tcpdump condition1 or condition2
```

Example:

```
tcpdump udp or icmp
```

Captures:

- UDP traffic
- ICMP traffic

NOT Operator

Captures packets where condition is false.

Syntax:

```
tcpdump not <condition>
```

Example:

```
tcpdump not tcp
```

Captures:

- UDP
- ICMP
- ARP
- Other non-TCP packets

Quick Command Cheat Sheet

`tcpdump host IP`	Filter by host
`tcpdump src host IP`	Filter source host
`tcpdump dst host IP`	Filter destination host
`tcpdump port 53`	Filter by port
`tcpdump src port 80`	Source port filter
`tcpdump dst port 443`	Destination port filter
`tcpdump icmp`	Protocol filter
`tcpdump udp or icmp`	Logical OR
`tcpdump host 1.1.1.1 and tcp`	Logical AND

Packet Filtering

Packet Length Filters

These filters display packets based on packet size.

1. greater LENGTH

Shows packets greater than or equal to the specified length.

Example:

```
tcpdump -r traffic.pcap 'greater 1500'
```

2. less LENGTH

Shows packets less than or equal to the specified length.

Example:

```
tcpdump -r traffic.pcap 'less 500'
```

TCP Flag Filtering

The TCP flags field can be accessed using:

```
tcp[tcpflags]
```

Common TCP Flags

tcp-syn	Synchronize
tcp-ack	Acknowledge
tcp-fin	Finish
tcp-rst	Reset
tcp-push	Push

Practical Examples

Count RST packets

```
tcpdump -r traffic.pcap 'tcp[tcpflags] == tcp-rst' | wc -l
```

Show SYN packets

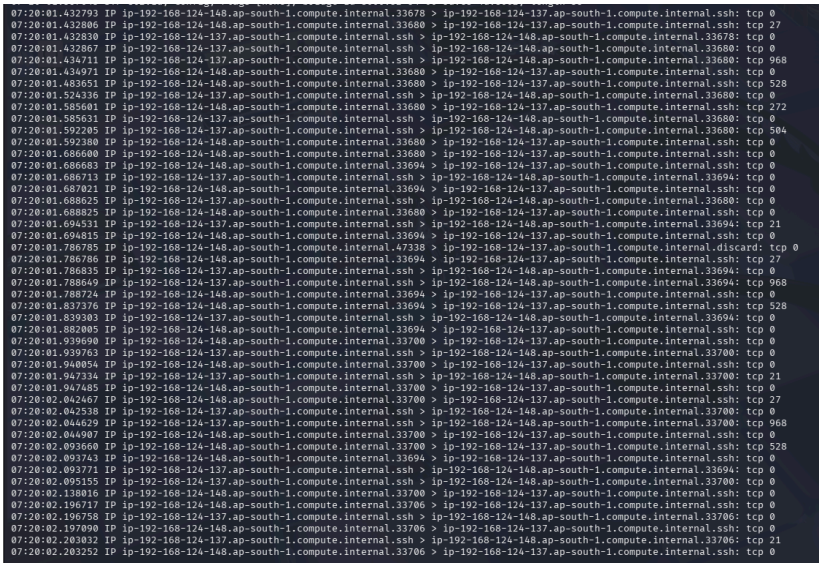
```
tcpdump -r traffic.pcap 'tcp[tcpflags] == tcp-syn'
```

Packets larger than 15000 bytes

```
tcpdump -r traffic.pcap 'greater 15000'
```

Output Formatting

Quick Output (-q)



Command:

```
tcpdump -r TwoPackets.pcap -q
```

Meaning:
Quick output displays brief packet information.

Displayed fields:

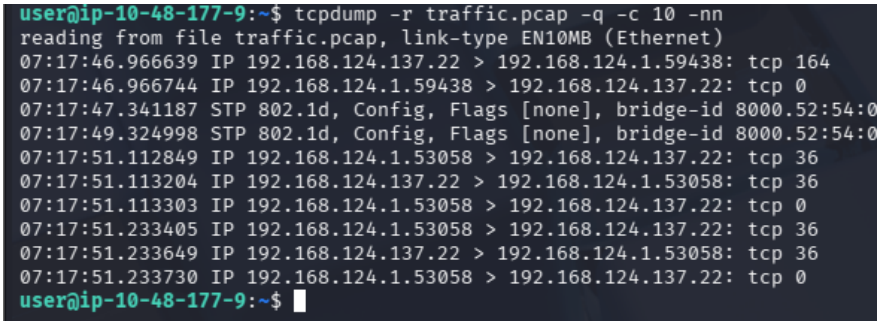
- Timestamp
- Source IP and port
- Destination IP and port
- Protocol
- Packet size

Display Link-Level Header (-e)

Command:

```
tcpdump -r TwoPackets.pcap -e
```

Adds link-layer header information.



Displayed fields:

- MAC address (source)
- MAC address (destination)
- Ethernet protocol type
- Packet length

Display Packet Data as ASCII (-A)

Command:

```
tcpdump -r TwoPackets.pcap -A
```

Meaning:
Displays packet payload as ASCII characters.

Display Packet Data in Hexadecimal (-xx)

Command:

```
tcpdump -r TwoPackets.pcap -xx
```

Displays packet contents byte-by-byte in hexadecimal format.

Example:
0x0000: 0283 1e40 5d17 44df 65d8 fe6c 0800 4500